

Pedigree analysis and relatedness inference

Magnus Dehli Vigeland and Thore Egeland

Solutions for exercise set IV: DIVIANA

Solution IV-1 (Plane crash)

- a) 8 victims, 5 missing persons and 5 families.
- b) Omitted.
- c) M2, M3 and M5 are *undisputedly* identified as V6, V2 and V4, respectively, while M1 has an inconclusive (but suggestive) match to V1.
- d) Let T denote the given LR threshold. A pairing $V_i = M_j$ is an *Undisputed* match if the corresponding LR is at least T , and at least T times larger than all other LRs involving V_i or M_j .
- e) All families are simple, with only one missing person, so no joint analysis was required.
- f) The previously *Inconclusive* match $M1 = V1$ is now *Probable*. This happens when the LR is below the threshold but larger than 10% of it.
- g) Omitted.
- h) V7 and V8 are clearly related, most likely as full siblings.

Solution IV-2 (Plane crash cont.)

- a) Individual 1 is *Nonidentifiable* (since she is unrelated to the reference R1).
- b) We now have a *Symmetric match* in family F2, since V6 matches both M2 and M2-2 equally well.
- c) A and B are both *Nonidentifiable*, since there are no typed relatives.

Solution IV-3 (Car crash)

- a) The dataset in the `.fam` file only specifies the mother as missing, not the two siblings. (Families can only handle families with a single missing person.) The labels are also slightly different.
- b) (Omitted.)
- c) *AM-PM*: The IBD estimates indicate that R1 is related to all of the PM samples. In particular, R1 and V1 are possible parent-child.

PM-PM: The estimates suggest that both V1-V2 and V1-V3 are parent-child relationships. The estimate for V2 and V3 is compatible with full sibs.
- d) The only exclusions are between M1 and V3, who have two incompatible markers.
- e) The LR matrix gives the result of the pairwise comparisons. We see that V1 gives high LR against both M1 and M3, while V2 shows a suggestive match against M2. The lack of clear conclusions motivates joint analysis.
- f) The GLR for $M2 = V2$ is $5.12e+11$. This is the LR comparing the first row (where $M2 = V2$) and second row (where $M2 \neq V2$) in the `Joint` table.
- g) When sex is ignored, the siblings M2 and M3 cannot be told apart, and are only identified as a pair. This is shown in DIVIANA as a *Symmetric match*. On a more practical note, ignoring sex nearly doubled the number of joint assignments, from 14 to 27.
- h) Only minor changes; the identifications appear robust.